Assignment 3: Longitudinal Text Messaging Analysis

Justin Coles, Nicole Bellantoni

November 11, 2026

Section 1: Environment Setup & Data Preparation

First, I loaded the `pastecs`, `tidyr`, and `dplyr` packages to perform data wrangling and descriptive statistics. Then, I imported `TextMessages.csv` from the project repository.

I converted `Group` into a labeled factor and the dataset into a tidy long format using `pivot_longer()` so that `Time` is a categorical variable with `Group`. Then, I exported this clean long dataset as `TextMessages_Clean_Long.csv` so my partner can import it directly to create the required faceted boxplots and bar charts.

```
# I loaded pastecs to compute detailed descriptive statistics using stat.desc()
library(pastecs)

# I loaded tidyr and dplyr to perform data wrangling and reshape the dataset
library(tidyr)
```

```
##
## Attaching package: 'tidyr'
```

```
## The following object is masked from 'package:pastecs':
##
##     extract
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:pastecs':
##
##     first, last
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# I imported the shared dataset from the project repository into text_data
text_data <- read.csv("TextMessages.csv")

# I checked the initial wide structure and dimension of the raw data
str(text_data)
```

```
## 'data.frame':    50 obs. of  4 variables:
```

```
##  $ Group      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Baseline   : int  52 68 85 47 73 57 63 50 66 60 ...
##  $ Six_months : int  32 48 62 16 63 53 59 58 59 57 ...
##  $ Participant: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
head(text_data)
```

```
##   Group Baseline Six_months Participant
## 1     1       52         32           1
## 2     1       68         48           2
## 3     1       85         62           3
## 4     1       47         16           4
## 5     1       73         63           5
## 6     1       57         53           6
```

```
# I converted Group into a labeled factor for clean categorical identification
text_data$Group <- factor(text_data$Group,
                          levels = c(1, 2),
                          labels = c("Group 1", "Group 2"))

# I reshaped Baseline and Six_months from wide into tidy long format
text_long <- text_data %>%
  pivot_longer(
    cols = c(Baseline, Six_months),
    names_to = "Time",
    values_to = "Text_Messages"
  ) %>%
  mutate(
    Time = factor(Time,
                  levels = c("Baseline", "Six_months"),
                  labels = c("Baseline", "6 Months"))
  )

# I verified the structure of the reshaped long data frame
str(text_long)
```

```
## tibble [100 x 4] (S3: tbl_df/tbl/data.frame)
##  $ Group        : Factor w/ 2 levels "Group 1","Group 2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Participant  : int [1:100] 1 1 2 2 3 3 4 4 5 5 ...
##  $ Time         : Factor w/ 2 levels "Baseline","6 Months": 1 2 1 2 1 2 1 2 1 2 ...
##  $ Text_Messages: int [1:100] 52 32 68 48 85 62 47 16 73 63 ...
```

```
head(text_long)
```

```
## # A tibble: 6 x 4
##   Group   Participant Time     Text_Messages
##   <fct>         <int> <fct>            <int>
## 1 Group 1           1 Baseline            52
## 2 Group 1           1 6 Months            32
## 3 Group 1           2 Baseline            68
## 4 Group 1           2 6 Months            48
## 5 Group 1           3 Baseline            85
## 6 Group 1           3 6 Months            62
```

```
# I saved the clean long data frame so my partner can use it for visual tasks
write.csv(text_long, "TextMessages_Clean_Long.csv", row.names = FALSE)
```

Section 2: Stratified Summary Statistics by Group and Time

I calculated summary statistics for `Text_Messages` stratified jointly by `Group` and `Time` using `by()` paired with `stat.desc()` from the `pastecs` package.

```
# I computed stratified summary statistics across Group and Time using stat.desc
stratified_summary <- by(
  text_long$Text_Messages,
  list(text_long$Group, text_long$Time),
  stat.desc,
  basic = TRUE
)

# I displayed the stratified summary statistics for Group 1 and Group 2 across both time points
stratified_summary
```

```
## : Group 1
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##    25.000000     0.000000     0.000000    47.000000    85.000000    38.000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
##  1621.000000    64.000000    64.840000     2.135946     4.408377   114.056667
##      std.dev     coef.var
##    10.679732     0.164709
## ------------------------------------------------------------
## : Group 2
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   89.0000000   43.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1640.0000000   65.0000000   65.6000000    2.1671794    4.4728385  117.4166667
##      std.dev     coef.var
##   10.8358971    0.1651814
## ------------------------------------------------------------
## : Group 1
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000    9.0000000   78.0000000   69.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1324.0000000   58.0000000   52.9600000    3.2662313    6.7411700  266.7066667
##      std.dev     coef.var
##   16.3311563    0.3083678
## ------------------------------------------------------------
## : Group 2
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   79.0000000   33.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1546.0000000   62.0000000   61.8400000    1.8820910    3.8844450   88.5566667
##      std.dev     coef.var
##    9.4104552    0.1521742
```

Summary:

Group 1 Statistics: * **Baseline (n = 25):** Mean = 64.84, Median = 64.00, SD = 10.68, Variance = 114.06, Range = 38.00 (Minimum = 47.00, Maximum = 85.00), SE = 2.14. * **6 Months (n = 25):** Mean = 52.96, Median = 58.00, SD = 16.33, Variance = 266.71, Range = 69.00 (Minimum = 9.00, Maximum = 78.00), SE = 3.27.

Group 2 Statistics: * **Baseline (n = 25):** Mean = 65.60, Median = 65.00, SD = 10.84, Variance = 117.42, Range = 43.00 (Minimum = 46.00, Maximum = 89.00), SE = 2.17. * **6 Months (n = 25):** Mean = 61.84, Median = 62.00, SD = 9.41, Variance = 88.56, Range = 33.00 (Minimum = 46.00, Maximum = 79.00), SE = 1.88.

Inferences:

1. **Baseline:** At Baseline, both intervention groups exhibited nearly identical messaging rates (Group 1 Mean = 64.84 vs. Group 2 Mean = 65.60) and very similar variability (SD = 10.68 vs. 10.84), showing that the groups were comparable prior to intervention.
2. **Longitudinal Change:** Over the 6-month period, Group 1 demonstrated a substantial decrease in text messaging, dropping by an average of 11.88 messages (from 64.84 down to 52.96). In contrast, Group 2 showed a much smaller reduction of only 3.76 messages (from 65.60 down to 61.84).
3. **Variance:** While Group 1 experienced a greater reduction in text messages, its spread widened considerably at 6 months (SD increased from 10.68 to 16.33; range increased from 38 to 69). In contrast, Group 2 maintained a tight, consistent spread (SD decreased from 10.84 to 9.41; range decreased from 43 to 33).